

System Information & Network Testing

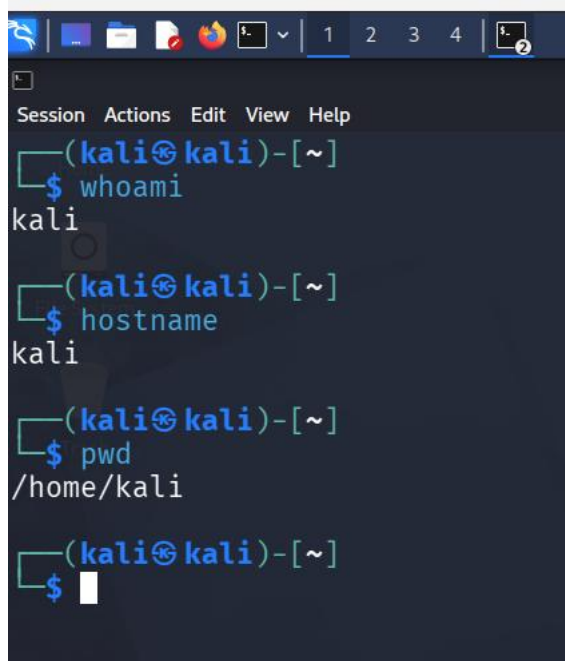
By

OGUNNAIKE Gbeke Sola,

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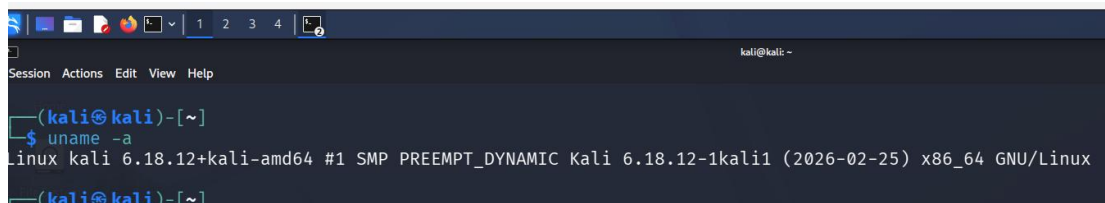
Deliverable: Step1

A terminal window with a dark background and light blue text. The window has a title bar with icons for file manager, web browser, and terminal. Below the title bar is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows four commands being executed: 'whoami' returns 'kali', 'hostname' returns 'kali', and 'pwd' returns '/home/kali'. The prompt is '(kali@kali)-[~]' for each command.

```
(kali@kali)-[~]  
$ whoami  
kali  
  
(kali@kali)-[~]  
$ hostname  
kali  
  
(kali@kali)-[~]  
$ pwd  
/home/kali  
  
(kali@kali)-[~]  
$
```

- The username shown by whoami is kali
- My machine hostname is kali
- The present working directory is /home/kali

Step 2 Operating System and Hardware Information

A terminal window showing the output of the 'uname -a' command. The window has a title bar with icons for file manager, web browser, and terminal. Below the title bar is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows the command 'uname -a' being executed, which returns 'Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux'. The prompt is '(kali@kali)-[~]' for each command.

```
(kali@kali)-[~]  
$ uname -a  
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64 GNU/Linux  
  
(kali@kali)-[~]  
$
```

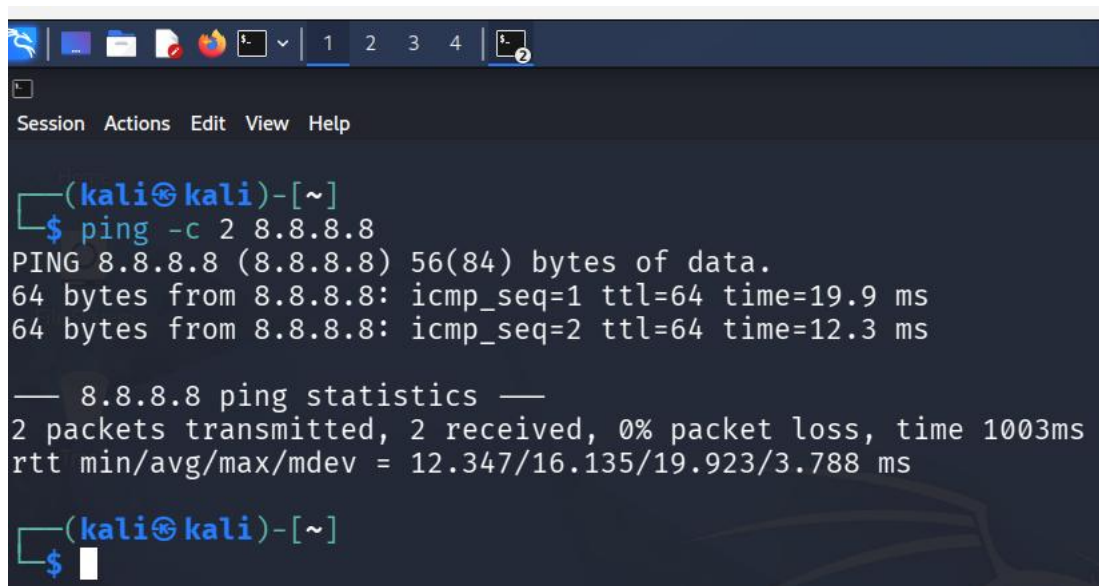
- My Linux kernel version is 6.18.12 amd64.

```
(kali㉿kali)-[~]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi         912Mi         225Mi         25Mi         1.0Gi         1.0Gi
Swap:          953Mi          12Ki         953Mi

(kali㉿kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            883M    0  883M   0% /dev
tmpfs           198M  992K   197M   1% /run
/dev/sda1       79G   16G   60G  21% /
tmpfs           986M   4.0K   986M   1% /dev/shm
none            1.0M    0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           986M  184K   986M   1% /tmp
none            1.0M    0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  116K   198M   1% /run/user/1000
```

- Total RAM of my Linux machine is 1.9Gi
- The disk space available on the main partition is 60G

Part 2 Basic Network Connectivity Test



```
(kali㉿kali)-[~]
$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=19.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=12.3 ms

— 8.8.8.8 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 12.347/16.135/19.923/3.788 ms

(kali㉿kali)-[~]
$
```

- The average round-trip time of ping -c 2 8.8.8.8 is 16.135ms
- There is no packet loss

```
(kali㉿kali)-[~]  
$ ping -c 4 google.com  
PING google.com (142.250.140.100) 56(84) bytes of data.  
64 bytes from wj-in-f100.1e100.net (142.250.140.100): icmp_seq=1 ttl=64 time=12.4 ms  
64 bytes from wj-in-f100.1e100.net (142.250.140.100): icmp_seq=2 ttl=64 time=11.0 ms  
64 bytes from wj-in-f100.1e100.net (142.250.140.100): icmp_seq=3 ttl=64 time=10.3 ms  
64 bytes from wj-in-f100.1e100.net (142.250.140.100): icmp_seq=4 ttl=64 time=11.4 ms  
  
— google.com ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3006ms  
rtt min/avg/max/mdev = 10.309/11.275/12.418/0.768 ms  
  
(kali㉿kali)-[~]  
$
```

- The resolved IP address of Google is 142.250.140.100
- The average rtt of ping google.com with -c 4 was smaller than rtt of 8.8.8.8 with -c 2. This is because google.com server might be physically closer to me than 8.8.8.8 server, therefore having a smaller rtt. Another reason could be due to different routing path for example, my internet services provider may have a better peering with google.com than google public DNS server (8.8.8.8)
- simply means "send 4 ping packets"

Exploring Other Linux Commands

Listing Directories and exploring Man page

```
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
Session Actions Edit View Help

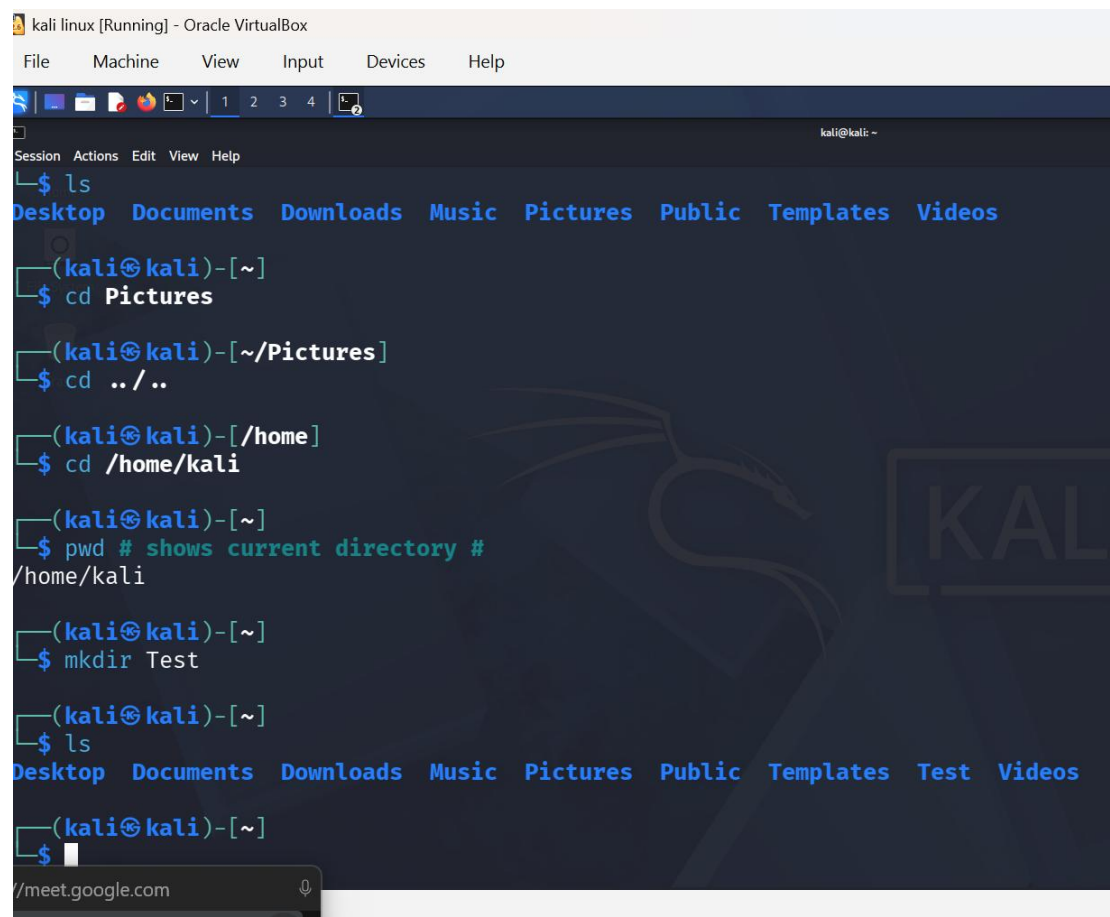
(kali@kali)-[~]
$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos

(kali@kali)-[~]
$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Desktop
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Documents
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Downloads
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Music
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Pictures
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Public
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Templates
drwxr-xr-x 2 kali kali 4096 Apr 27 17:19 Videos

(kali@kali)-[~]
$ man ls

(kali@kali)-[~]
$
```

Changing and creating directories



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos

(kali@kali)-[~]
$ cd Pictures

(kali@kali)-[~/Pictures]
$ cd ../../

(kali@kali)-[/home]
$ cd /home/kali

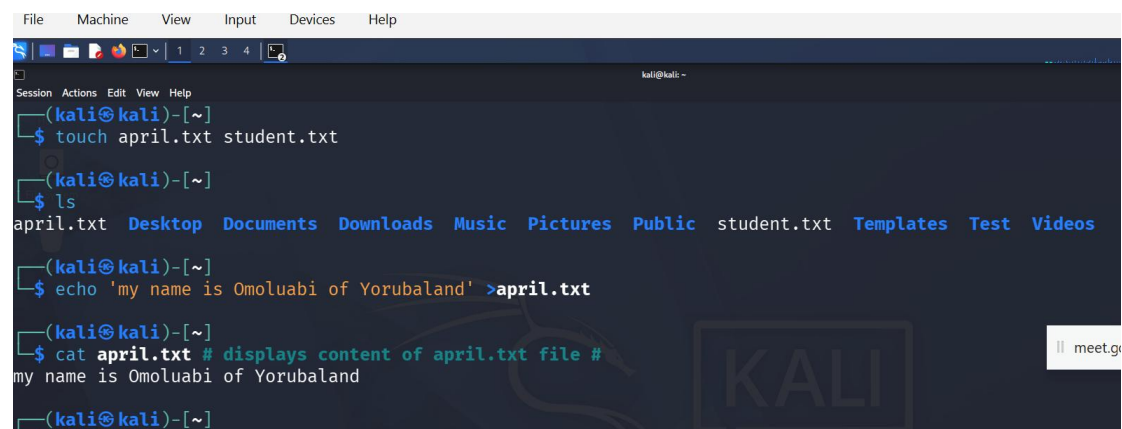
(kali@kali)-[~]
$ pwd # shows current directory #
/home/kali

(kali@kali)-[~]
$ mkdir Test

(kali@kali)-[~]
$ ls
Desktop Documents Downloads Music Pictures Public Templates Test Videos

(kali@kali)-[~]
$
```

Creating and copying content file



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
$ touch april.txt student.txt

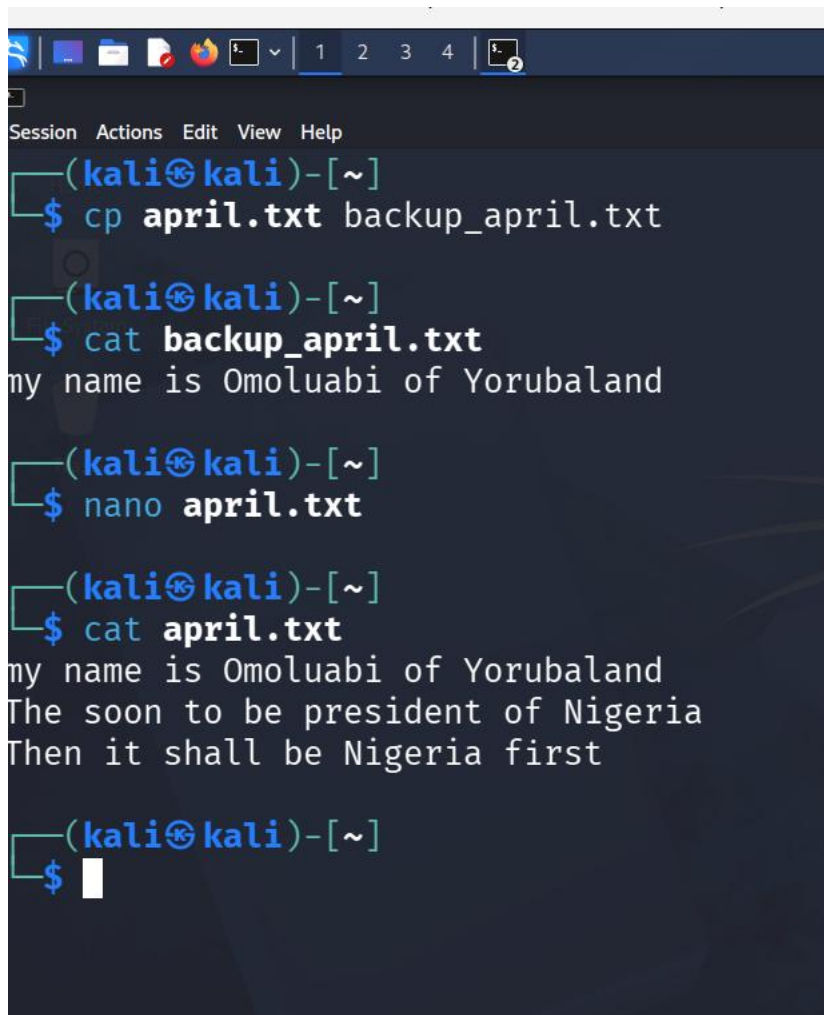
(kali@kali)-[~]
$ ls
april.txt Desktop Documents Downloads Music Pictures Public student.txt Templates Test Videos

(kali@kali)-[~]
$ echo 'my name is Omoluabi of Yorubaland' >april.txt

(kali@kali)-[~]
$ cat april.txt # displays content of april.txt file #
my name is Omoluabi of Yorubaland

(kali@kali)-[~]
$
```

Copy and editing file content



A terminal window on a Kali Linux system. The prompt is `(kali㉿kali)-[~]`. The user runs `$ cp april.txt backup_april.txt`. Then `$ cat backup_april.txt` displays the text: `my name is Omoluabi of Yorubaland`. Next, the user runs `$ nano april.txt`. Then `$ cat april.txt` displays the text: `my name is Omoluabi of Yorubaland`, `The soon to be president of Nigeria`, and `Then it shall be Nigeria first`. The prompt `(kali㉿kali)-[~]` is shown again with a cursor on the next line.

```
(kali㉿kali)-[~]
$ cp april.txt backup_april.txt

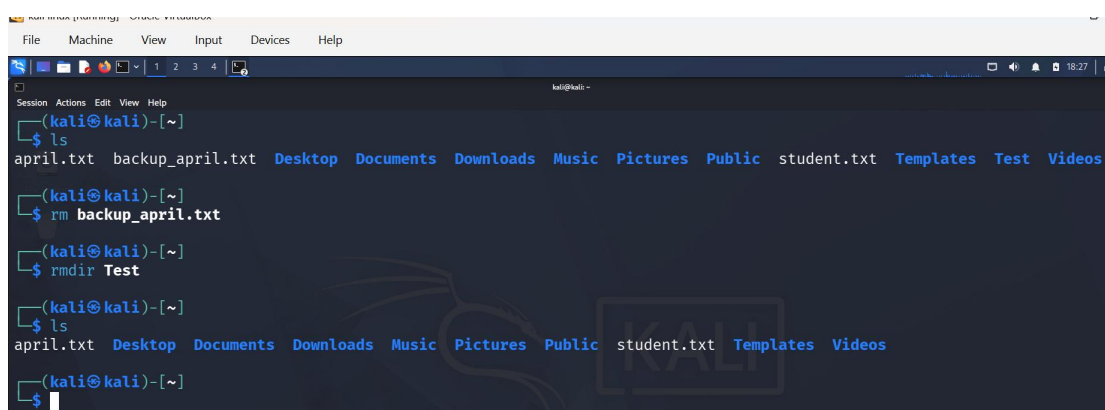
(kali㉿kali)-[~]
$ cat backup_april.txt
my name is Omoluabi of Yorubaland

(kali㉿kali)-[~]
$ nano april.txt

(kali㉿kali)-[~]
$ cat april.txt
my name is Omoluabi of Yorubaland
The soon to be president of Nigeria
Then it shall be Nigeria first

(kali㉿kali)-[~]
$
```

Removing files and directories



A terminal window on a Kali Linux system. The prompt is `(kali㉿kali)-[~]`. The user runs `$ ls`, showing the file list: `april.txt backup_april.txt Desktop Documents Downloads Music Pictures Public student.txt Templates Test Videos`. Then the user runs `$ rm backup_april.txt`. Then `$ rmdir Test`. Then `$ ls` shows the updated file list: `april.txt Desktop Documents Downloads Music Pictures Public student.txt Templates Videos`. The prompt `(kali㉿kali)-[~]` is shown again with a cursor on the next line.

```
(kali㉿kali)-[~]
$ ls
april.txt backup_april.txt Desktop Documents Downloads Music Pictures Public student.txt Templates Test Videos

(kali㉿kali)-[~]
$ rm backup_april.txt

(kali㉿kali)-[~]
$ rmdir Test

(kali㉿kali)-[~]
$ ls
april.txt Desktop Documents Downloads Music Pictures Public student.txt Templates Videos

(kali㉿kali)-[~]
$
```

Moving files into Directory

```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ ls
april.txt Desktop Documents Downloads Music Pictures Public student.txt Templates Videos

(kali@kali)-[~]
$ mv student.txt Desktop

(kali@kali)-[~]
$ cd Desktop

(kali@kali)-[~/Desktop]
$ ls
student.txt

(kali@kali)-[~/Desktop]
$
```

Less command

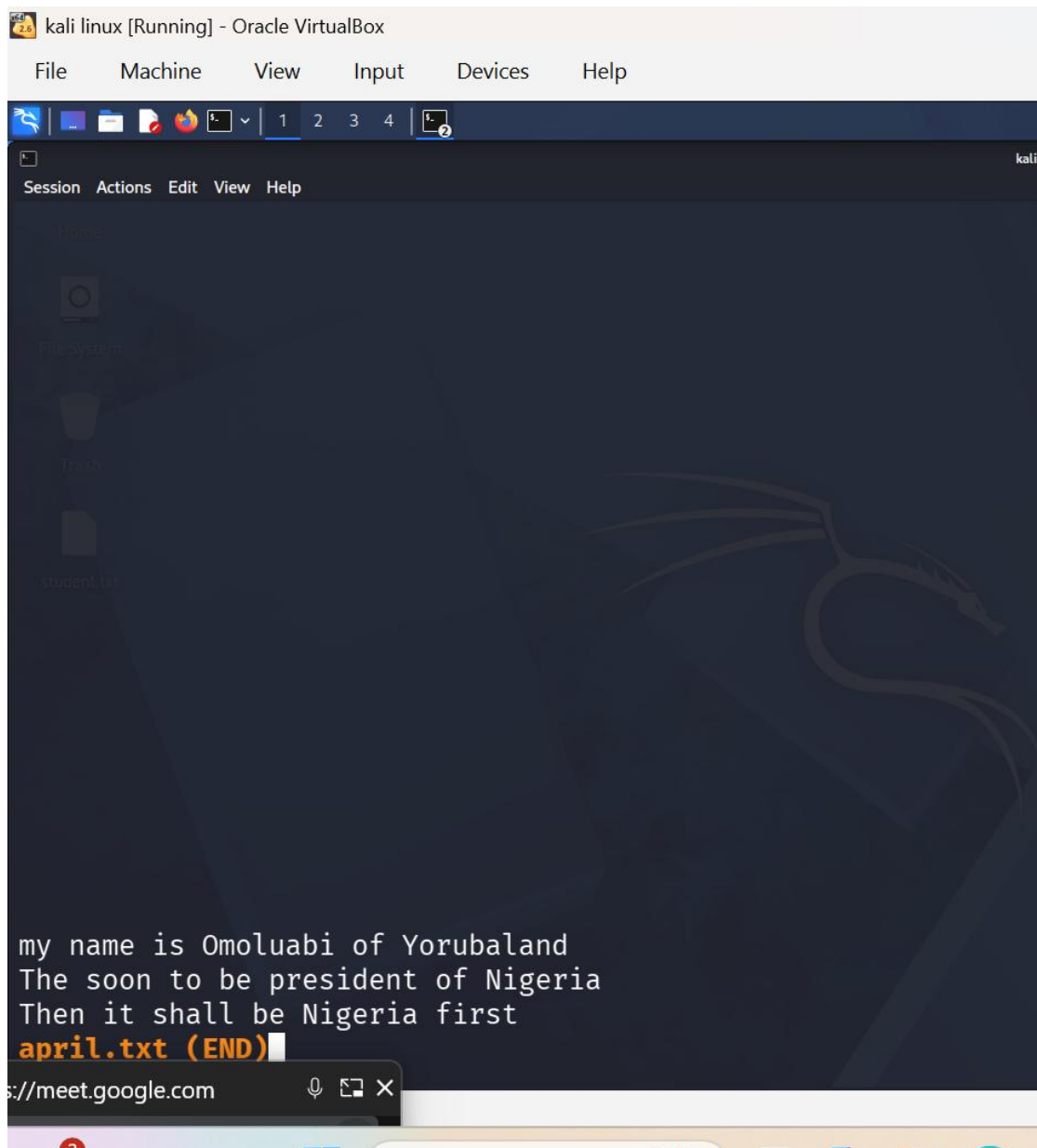
```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~/Desktop]
$ cd /home/kali

(kali@kali)-[~]
$ cat april.txt
my name is Omoluabi of Yorubaland
The soon to be president of Nigeria
Then it shall be Nigeria first

(kali@kali)-[~]
$ less april.txt

(kali@kali)-[~]
$
```



Installing Updates and APT

```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/non-free amd64 Packages [189 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [908 kB]
Fetched 75.7 MB in 25s (2,992 kB/s)
1128 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)-[~]
$
```

```
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo apt install curl
The following packages were automatically installed and are no longer required:
base58 python3-browser-cookie3 python3-pyfiglet
binutils-mingw-w64-base python3-cbor python3-pyotp
binutils-mingw-w64-i686 python3-celery python3-pyqrcode
binutils-mingw-w64-x86-64 python3-cli-helpers python3-pyramid
debugedit python3-click-didyoumean python3-pyshodan
dnsmap python3-click-repl python3-qasync
dsniff python3-cmd2 python3-qrcode
ettercap-common python3-configobj python3-serial-asyncio
ettercap-graphical python3-cpuinfo python3-sgmllib3k
faraday-agent-dispatcher python3-cvss python3-sh
figlet python3-ecdsa python3-shtab
finger python3-elastic-transport python3-simple-rest-client
gcc-mingw-w64-base python3-elasticsearch python3-simplekv
gcc-mingw-w64-i686-win32 python3-ephem python3-slugify
gcc-mingw-w64-i686-win32-runtime python3-faraday-agent-parameters-types python3-smmmap
gcc-mingw-w64-x86-64-win32 python3-faraday-plugins python3-snappy
gcc-mingw-w64-x86-64-win32-runtime python3-feedparser python3-sniffio
python3-filedescriptor python3-spinners
Installing:
curl
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1128
Download size: 277 kB
Space needed: 513 kB / 63.6 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 curl amd64 8.19.0-3 [277 kB]
Fetched 277 kB in 0s (736 kB/s)
Selecting previously unselected package curl.
(Reading database... 392392 files and directories currently installed.)
Preparing to unpack .../curl_8.19.0-3_amd64.deb...
Unpacking curl (8.19.0-3)...
Setting up curl (8.19.0-3)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...

(kali@kali)-[~]
$
```

```

(kali㉿kali)-[~]
$ curl --version #to that curl has been install#
curl 8.19.0 (x86_64-pc-linux-gnu) libcurl/8.19.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2
libssh2/1.11.1 nghttp2/1.68.0 ngtcp2/1.21.0 nghttp3/1.12.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-03-11, security patched: 8.19.0-3
Protocols: dict file ftp ftps gopher gophers http https imap imaps ipfs ipns ldap ldaps mqtt mqtts pop3 pop3s rtmp rtsp scp sftp
smbs smb smbssl smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTP3 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe
TLS-SRP UnixSockets zstd
(kali㉿kali)-[~]

```

Removing curl installation

```

(kali㉿kali)-[~]
$ sudo apt remove curl
The following packages were automatically installed and are no longer required:
base58 python3-browser-cookie3 python3-pyfiglet
binutils-mingw-w64-base python3-cbor python3-pyotp
binutils-mingw-w64-i686 python3-celery python3-pyqrcode
binutils-mingw-w64-x86-64 python3-cli-helpers python3-pyramid
debugedit python3-click-didyoumean python3-pyshodan
dnsmmap python3-click-repl python3-qasync
dsniff python3-cmd2 python3-qrcode
ettercap-common python3-configobj python3-serial-asyncio
ettercap-graphical python3-cpuinfo python3-sgmllib3k
faraday-agent-dispatcher python3-cvss python3-sh
figlet python3-ecdsa python3-shtab
finger python3-elastic-transport python3-simple-rest-client
gcc-mingw-w64-base python3-elasticsearch python3-simplekv
gcc-mingw-w64-i686-win32 python3-ephem python3-slugify
gcc-mingw-w64-i686-win32-runtime python3-faraday-agent-parameters-types python3-smmmap
gcc-mingw-w64-x86-64-win32 python3-faraday-plugins python3-snappy
gcc-mingw-w64-x86-64-win32-runtime python3-feedparser python3-sniffio

REMOVING:
curl

Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1128
Freed space: 513 kB

Continue? [Y/n] y
(Reading database... 392404 files and directories currently installed.)
Removing curl (8.19.0-3)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...

(kali㉿kali)-[~]
$

```